

6 Chapter 6

6.1 Review

Wire geometry, resistance, capacitance, inductance. π -model for finding Elmore delay. Use sheet resistance to eliminate detail and find resistance per unit length, using the width. Scaling affects RC wire delay. Crosstalk delay, Miller effect (**note:** definitely look further into Miller effect.) Capacitor voltage divider. Repeaters! Best length of wire between inverters, best repeater size.

6.2 Questions

Pi-model question, resistivity question (sheet), multiple-choice scaling/RC question, Miller effect, maybe repeater.

7 Chapter 7

7.1 Review

Process variation can be rolled into one variation parameter, K_{process} , so we don't need to labor over the details. Note that K_{load} is given in $\frac{\text{ns}}{\text{pF}}$, so find the wire + gate capacitance of the load and multiply by K_{load} . Dennard scaling is the simplest assumption, where all dimensions scale in the same way to ensure a constant electric field.

7.2 Questions

TSMC cell library derating factors, effects of voltage, temperature, process (all possible manufacturing factors rolled into one). Might include loaded gates. Definitely a Dennard scaling question.

8 Chapter 9

8.1 Review

Inner/outer input issues (**review**). Why does this happen? Thinking about a series nMOS network, say B and C inputs are already 1 (the only condition in which a rising edge on A could cause a falling Y output). Generalize this to a NOR gate? Same result? No, but we might have a similar situation on the rising edge.

Say B transition causes Y to rise, then A must already be 0, and the A capacitor pre-charged. In the alternate case, where A transitions *last*, both capacitors will need to charge. Asymmetric gates are complex to analyze, but the basic rationale makes sense. Skewed gates, **review**. Gets away from matching pull-up and pull-down strengths (why would you do this?) Ratioed circuits use an always-on pMOS transistor (basically a resistor). Key feature is the ratio of pull-up current to pull-down current (**review**). F e e t. Monotonic inputs, you can repeat these to make them work, but you can't cascade *dynamic* gates.

8.2 Questions

Three-input multiple choice, NAND/NOR. Fill in transistor sizing for HI-skew and LO-skew (multiple-choice?). Probably something on ratioed and/or f e e t.